NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER 2 SOLUTIONS.

SATURDAY 28TH JUNE, 2025 9:00 AM - 12:15 PM

SECTION A: 5 marks each

1. Let $\frac{x-y}{x+y} = \frac{2024}{2025}$. What is the value of $\frac{x}{y}$?

Solution

Consider $\frac{x-y}{x+y} = \frac{2024}{2025} \Rightarrow \frac{\frac{x}{y}-1}{\frac{x}{y}+1} = \frac{2024}{2025}$, solving for $\frac{x}{y}$ as avariable gives $\frac{x}{y} = 4049$.

2. In a certain town, five people were engaged in a heated debate as follows;

   Hosea: Betty, the probability that you win in a UMS contest is 50% because it is either you win or you lose therefore your chances of winning is $50\% + 50\% = 100\%$ if you participate in paper I and Paper II so participate in paper I and II and win

   Margaret: Hosea, you are smart

   Betty: Cool! based on Hosea's statement, I will win one of the contests

   Lisa: You are wrong

   Berna: All of you are wrong

   Who among the five is right?

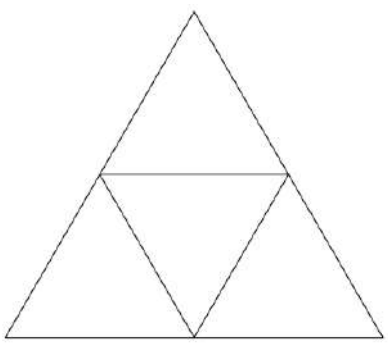
Solution

Lisa is right since all the people's statement above her are wrong.

3. Prove that any five points inside an equilateral triangle of side length 2, there are two pints whose distance is at most 1.

Solution

We divide the equilateral triangle into **four** congruent (similar) parts, as shown in the figure below:



Then there must be two points out of the five that belong to the same part. The distance between those two points is clearly at most the side length of the smaller equilateral triangles, which is 1.

4. Let $f(x) = \frac{1}{1+2x}$. What is the sum of all numbers x satisfying $f(f(x)) = x$?

Solution

$f(f(x)) = x \Rightarrow f(f(x)) = f\left(\frac{1}{1+2x}\right) = \frac{1+2x}{3+2x}$, now $\frac{1+2x}{3+2x} = x \Rightarrow x = -1, \frac{1}{2}$ thus $\frac{1}{2} + -1 = -\frac{1}{2}$.

5. How many ways are there to fill in cells of a 3×3 empty table with 2 or -2 such that the product in each row or column equals to 8, for example

-2	2	-2
2	2	2
-2	2	-2

Solution

16 ways (one can list all ways)

6. We call a positive integer *powerfree* if none of its digits can be written as a power of an integer with an exponent bigger than 1 for example the number 53 is *powerfree* but 54 is not *powerfree* since $4 = 2^2$. Find the difference between the largest and the smallest two digit *powerfree* numbers?

Solution

Smallest two digit power free number is 22 and the largest is 77 thus the difference is $77 - 22 = 55$ [Use the digits $0, 1, 2, 3, 4, \cdots, 9$ to eliminate powerfree numbers].

7. Let f be a function with the following properties;

   (a) $f(1) = 1$
   (b) $f(2n) = n \cdot f(n)$

   What is the value of $f(2^9)$?

Solution

$f(2^9) = f(2 \cdot 2^8) = 2^8 \cdot f(2^8) = 2^8 \cdot 2^7 \cdot f(2^6)$ continuing this way we get $f(2^9) = 2^{8+7+6+5+4+3+2+1} f(2^0)$ but $f(1) = 1$, therefore $f(2^9) = 2^{36}$.

8. Suppose that we choose an integer from 1 to 400 (inclusive), where we have equal probability of selecting any integer in the range. What is the probability that we select an even integer which has a 3 as one of its digits?

Solution

we have 400 numbers in the range 80 sample space is 400. For an integer to have a 3 as one of its digits and is even then one of the two options are possible;

(a) 3 in tens position i.e – 3 – implying $3 \times 1 \times 5 = 15$ ways

(b) 3 in hundreds position i.e 3—— implying $10 \times 5 = 50$ ways, total ways $= 50 + 15 = 65$ and therefore the required probability is $\frac{65}{400} = \frac{13}{80}$.

9. Suppose two armies at war are advancing towards each other, each one at $1km/h$. A neutral messenger(not at war with both armies) leaves the first army when the two armies are $10km$ apart and runs towards the second at $9km/h$. Upon reaching the second army, he immediately turns around and runs towards the first army at $9km/h$. How many miles apart are the two advancing armies when the messenger gets back to the first army?

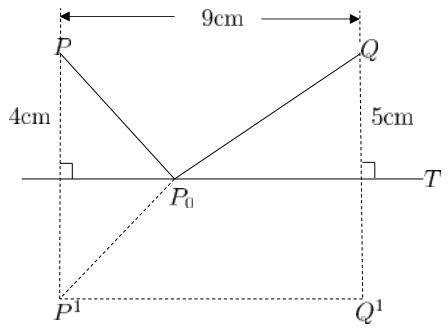
Solution

The messenger reaches the second army in one hour (since the messenger approaches the second army at a combined speed of $10km/h$) so the time taken $= \frac{distance}{speed} = \frac{10km}{10km/h} = 1hr$. At that time the two armies are $8km$ apart. It takes $0.8hrs$ for the messenger to get back to the first army since distance = 8km and speed is 10km/h. The armies would be $(8 - 0.8 - 0.8)km = 6.4km$ apart.

10. Two points P and Q are vertically 4 cm and 5 cm away from the line T, respectively and they are horizontally 9 cm apart. When a point P_0 moves along the line T. What is the smallest value of $AP_0 + P_0B$?

Solution

Let P' and Q' be the reflections of P and Q with respect to T and let P_0 be the point at which $P'Q$ meets T. The sum $PP_0 + P_0Q$ minimizes at P_0



From the sketch above, we have $PP_0 + P_0Q = P'Q = \sqrt{(PQ)^2 + (QQ')} = \sqrt{162}$ units.

SECTION B: 10 marks each

11. Let a and b be two real numbers satisfying the equation

$$ab + \sqrt{ab+1} + (\sqrt{a^2+b})(\sqrt{b^2+a}) = 0.$$

Determine the value of $a\sqrt{b^2+a} + b\sqrt{a^2+b}$.

Solution

- Let us rewrite the given equation as follows:

$$ab + (\sqrt{a^2+b})(\sqrt{b^2+a}) = -\sqrt{ab+1}.$$

- Now, squaring both sides of the above equation yields:

$$a^2b^2 + 2ab(\sqrt{a^2+b})(\sqrt{b^2+a}) + (a^2+b)(b^2+a) = ab+1.$$

- After some rearranging the preceeding equation becomes:

$$(a^2b^2 + a^3) + 2ab(\sqrt{a^2+b})(\sqrt{b^2+a}) + (a^2b^2+b^3) = 1.$$

- This simplifies to:

$$(a\sqrt{b^2+a} + b\sqrt{a^2+b})^2 = 1 \quad \Rightarrow$$
$$a\sqrt{b^2+a} + b\sqrt{a^2+b} = \pm 1.$$

- Next, we show that $a\sqrt{b^2+a} + b\sqrt{a^2+b} > 0$. Note that $ab = -\sqrt{ab+1} - (\sqrt{a^2+b})(\sqrt{b^2+a}) < 0$, so a and b have opposite signs.
- Without loss of generality, we may assume that $a > 0 > b$. Then rewrite

$$a\sqrt{b^2+a} + b\sqrt{a^2+b} = a(\sqrt{b^2+a}+b) - b(a - \sqrt{a^2+b}).$$

- Since $\sqrt{b^2+a}+b$ and $a-\sqrt{a^2+b}$ are both positive, the expression on LHS of the preceeding equation is positive. Therefore, $\boxed{a\sqrt{b^2+a} + b\sqrt{a^2+b} = 1}$.

12. A fair six-sided die is rolled. What is the probability that the product of the five visible numbers is greater than 150?

Solution

Let k denote the invisible number of the rolled die,

- If $k = 6$ then the product of visible number is $1 \times 2 \times 3 \times 4 \times 5 = 120$

- If $k = 5$ then the product of visible number is $1 \times 2 \times 3 \times 4 \times 6 = 144$
- If $k = 4$ then the product of visible number is $1 \times 2 \times 3 \times 5 \times 6 = 180 > 150$
- If $k = 3$ then the product of visible number is $1 \times 2 \times 4 \times 5 \times 6 = 240 > 150$
- If $k = 2$ then the product of visible number is $1 \times 3 \times 4 \times 5 \times 6 = 360 > 150$
- If $k = 1$ then the product of visible number is $2 \times 3 \times 4 \times 5 \times 6 = 720 > 150$

so required $prob = \frac{1 \le k \le 4}{1 \le k \le 6} = \frac{4}{6} = \frac{2}{3}$

13. Find the number of integers between 1 and 2025 (inclusive) that share a prime factor with 10?

Solution

Prime factors of 10 are $\{2, 5\}$ now,

- all multiples of 2 share prime factor 2 and their number is the integer part of $\frac{2025}{2} = 1012$ that is multiples of 2 between 1 and 2025 that share prime factor 2.
- all multiples of 5 share prime factor 5 and their number is the integer part of $\frac{2025}{5} = 405$ that is multiples of 5 between 1 and 2025 that share prime factor 5.
- all integers not having either of the factors 2 or 5 do not share a common factor and they are 202 between 1 and 2025 e.g $10, 20, 30, \cdots$ thus those that share are $(1012 + 405) - 202 = 1215$ integers

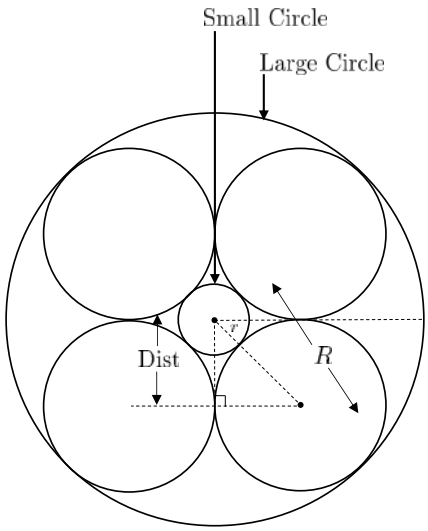
14. John's compound has twice as much area as Betty's Compound and three times as much as Caroline's compound. Caroline's Compound mower cuts half as fast as that of Betty's mower and one third as fast as John's mower. If they all start to mow their compounds at the same time who will finish first?

Solution

Let John's compound have an area A, then Betty's compound will have $\frac{A}{2}$ area and Caroline's will be $\frac{A}{3}$, Let John's compound mower cut at the rate of V, thus time for cutting the compound is $\frac{Area}{Speed}$ thus John's compound will be cut in $\frac{A}{V}$ time, Betty's compound will be cut in $\frac{3A}{4V}$ time and Caroline's compound to be cut in $\frac{A}{V}$ so Betty will be the first to mow her compound.

15. Four circles of radius $1cm$ each are arranged symmetrically so that each is tangent to two others. What is the ratio of the radius of the large circle to radius of the small circle that can be formed tangent to all the four circles.

Solution



Let r be the radius of the small circle and R be the radius of the large (outer circle). Then from the sketch above, $(r+1)^2 - dist^2 = 1^2$, thus $dist^2 = r^2 + 2r$. From $dist + 1 = 2$ then, $r = \sqrt{2} - 1$ and $R = r + 2 = \sqrt{2} + 1$.
Now the ratio required $\frac{R}{r} = \frac{\sqrt{2}+1}{\sqrt{2}-1.} = 3 + 2\sqrt{2} > 1$.

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